NLP for social Interactions

NLP and Identity

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First Assignment

Goals

- ✓ how to conduct a research project (small scale)
- ✓ practice to communicate results of experiments
- ✓ practical work / practice implementation, plots etc.
- ✓ practice interpreting results and scientific and critical thinking
- ✓ independent working, managing a longer project

First Assignment

Content

Option 1

Linguistic Variation

Investigate linguistic variation between communities and speakers

- pick 2 to 3 subreddits
- develop a hypothesis about which linguistic features or patterns could be interesting to investigate (if there is no clear hypothesis pick a set of features that you find interesting or intuitive to understand)
- extract the target features
- analyze the differences between the communities based on features or distinct words (e.g. by using plots or statistical analysis)
- train a feature-based classifier to predict which community a post comes from
- **optional**: investigate speaker-based differences

Research Questions

- → How does the style of community A vary compared to community B?
- → How well can [my features] predict which community a post comes from?
- → Which features are most informative in the prediction?
- → How much variation do we find between speakers in one community? (is it a homogenous community?)

Data

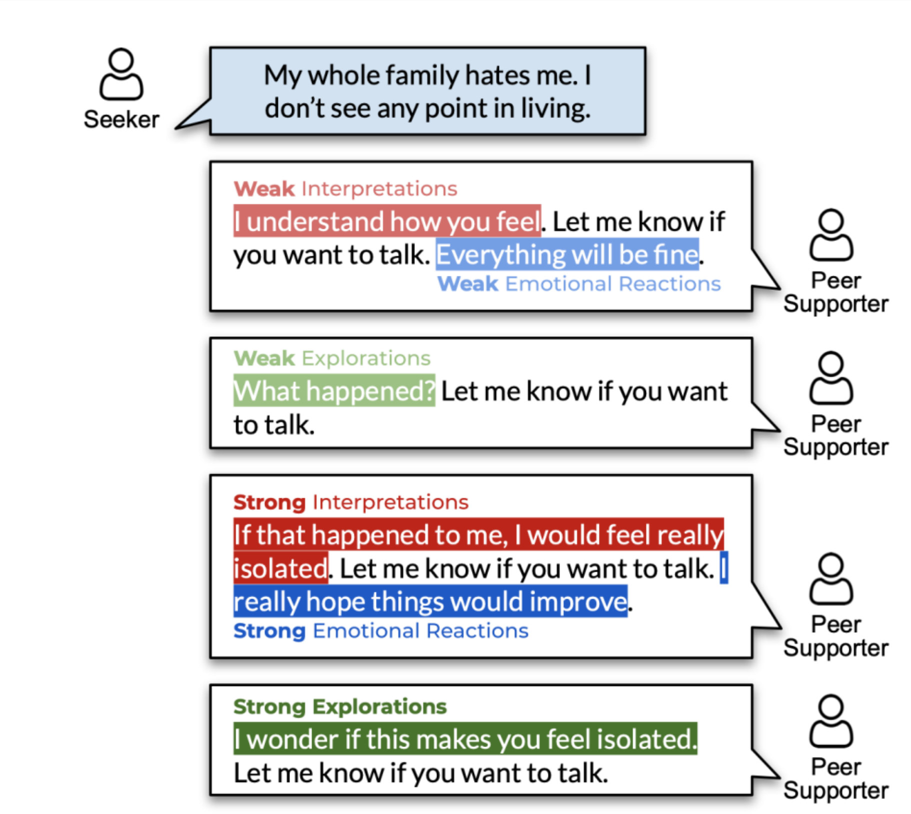
reddit: all subreddits available in convokit can be found here:

https://zissou.infosci.cornell.edu/convokit/datasets/subreddit-corpus/corpus-zipped/

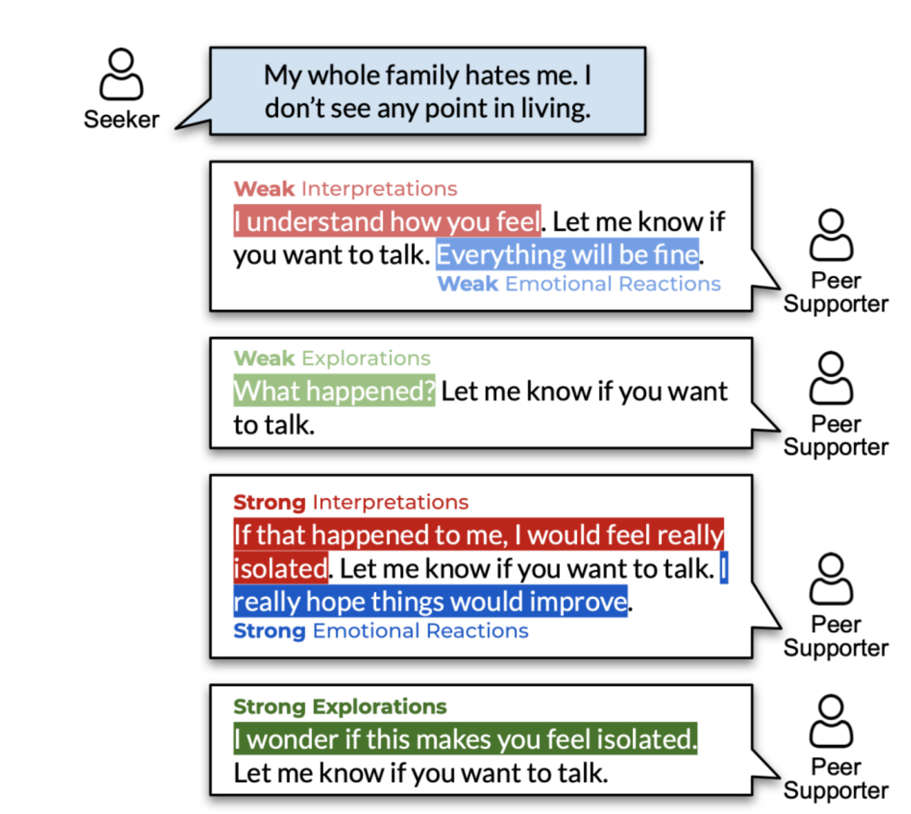
features: you can use any feature extraction tool you like (e.g. lftk, spacy, politeness strategies…)

Option 2

Modeling Empathy



Modeling Empathy
EPITOME Dataset



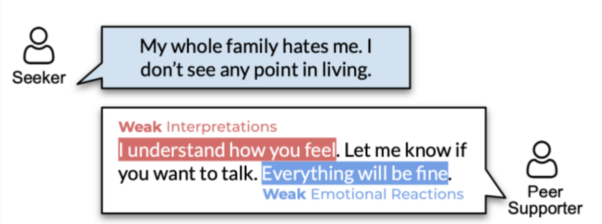
Emotional Reactions: Expressing emotions such as warmth, compassion, and concern, experienced by peer supporter after reading seekers post

Interpretations: Communicating an understanding of feelings and experiences inferred from the seekers post.

Explorations: Improving understanding of the seeker by exploring the feelings and experiences not stated in the post. Showing an active interest in what the seeker is experiencing and feeling

link to the paper: https://aclanthology.org/2020.emnlp-main.425.pdf

Modeling Empathy



- pick two types of empathy (from the three)
- train two feature-based classifier for each of them: one should use some linguistic features (e.g. extracted with lftk, politeness strategies), one should be a bag-of-words classifier. Investigate how well they do on the task.
- investigate which features or words correlated with which type of empathy
- build your own small test set with post-response pairs (20). They can come from any existing dataset / be written by yourself or generated. Annotate them for one type of empathy. Use the classifiers to predict on that test set and analyze the results. Which instances are missed?
- **optional**: add a BERT classifier and compare the results

Research Questions

- How well can linguistic features predict the annotated type of empathy?
- How well can a simple BoW classifier predict the annotated type of empathy?
- With which features / words does exploration / interpretation / emotional reaction correlate?
- Can the model accurately model empathy on new data? Which instances does it miss?

Data

https://github.com/behavioral-data/Empathy-Mental-Health

A: “it's too much. i can't take it anymore”
B: “You aren't alone here. What's going on?”
label: 1 (weak communication of exploration)

Formalities

Minimum Requirements

- → Option 1: analysis of differences + results of feature-based classifier + report
- → Option 2: results of the models + annotation and analysis + report

- → report: the template with descriptions can be found here: https://www.overleaf.com/read/frrrhwzsnrpf#b87098 [needs to be fixed]
- → teamwork: you are allowed to work in pairs on the project (implementation, analysis of results) but each person has to write their own report

Timeline

Date	
10.12	decide for one option (and subreddit if option 1 and the features you want to investigate). Formulate your research questions, get familiar with the data
17.12	work on the feature extraction / modeling
23.01	submission deadline, report + code

Server

- → create your own directory under /mount/studenten-temp1/users
- → an additional document with helpful information on the server will be uploaded

Report

- abstract: summary of the project
- introduction: motivation and research questions
- experimental setup: what methods do you use? what data do you use? hyperparameters? what features do you extract? what do the features mean?
- analysis: with sub-sections for each sub-topic. answer the research questions
- discussion: what errors do you find in the modeling? what ethical issues do you see with the task? what issues did you find when you did the task?

GenAI

- limit your use of generative AI in the writing
- use *your own words*
- the quality of the language is not graded, the graded part is the content. Critical thinking and making your own conclusions and observations is graded (GenAI tendentially obscures your own thoughts and sounds very generic)
- avoid LLM expressions (”delve deep”, ...)